

5. (a) Smart materials are used to make the frames and lenses of certain spectacles.



Give the names of the different types of smart material used. Describe the unusual property of each. [2]

Frames

.....

.....

Lenses

.....

.....



(b) Titanium dioxide has been used in sun creams for decades. Sun screens available today use nano-scale titanium dioxide particles. Some people believe using creams containing nano-particles is unsafe.

(i) Give the advantage of using nano-scale titanium dioxide particles rather than larger titanium dioxide particles to make sun screens. [1]

.....

.....

(ii) Explain why some people are concerned about the use of nano-scale titanium dioxide particles in sun screens. [2]

.....

.....

.....

(iii) **Approximately** how many times bigger are common titanium dioxide particles ($3 \times 10^{-7} \text{ m}$) than nano-scale titanium dioxide particles ($2.5 \times 10^{-10} \text{ m}$)? [2]

Answer =

